

Kello User Guide for CM

Revision 1.0

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Document information

Abstract

This document describes how to use Kello prototype

Document convention

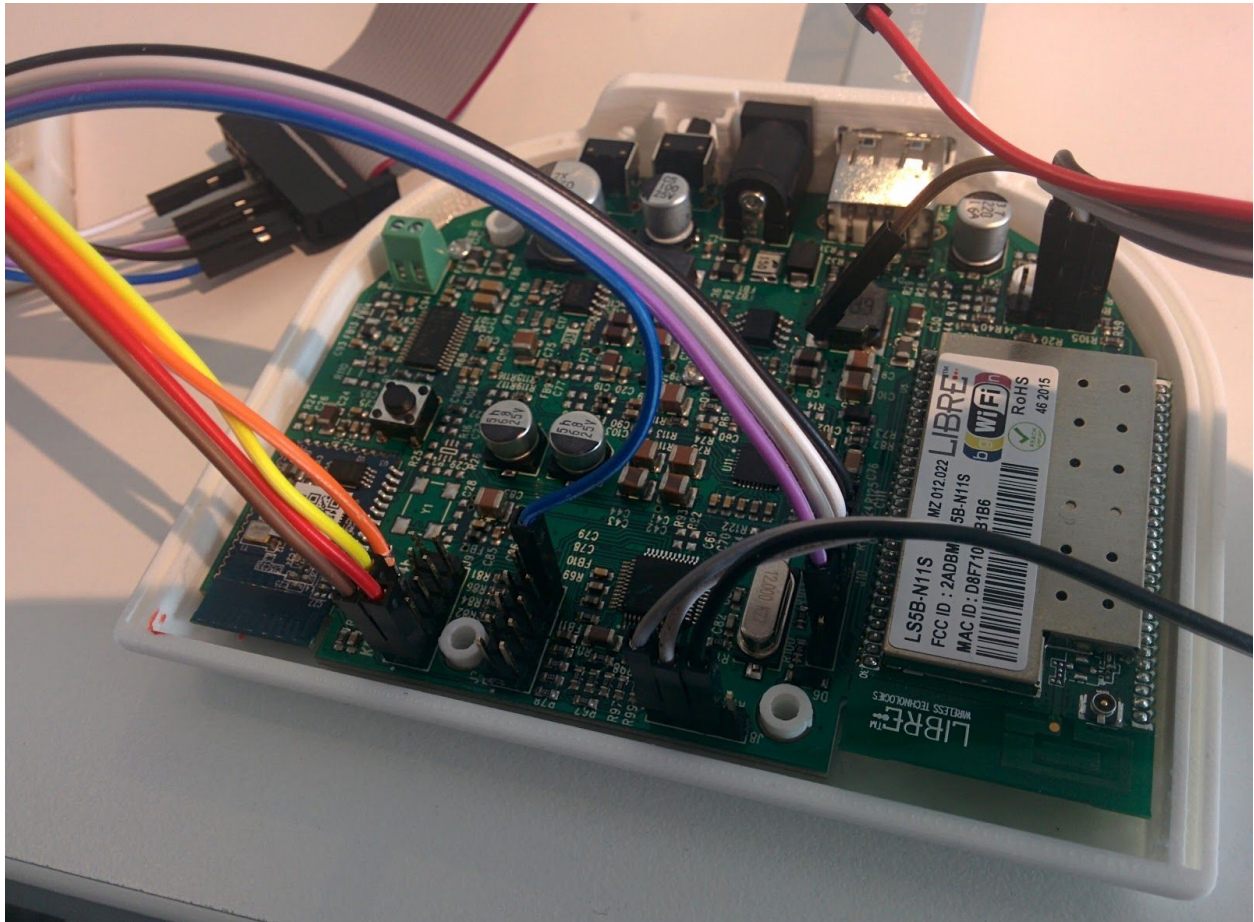
Revision history

Revision	Date	Description	Author
1.0	02/12/2016	Initial revision	Frederic Germain

Introduction

This document describes how to use Kello prototype

Kello1 board



white/grey/black front: serial debug rx/tx/gnd mcu 115200 8N1
Grey/white/black back: serial debug rx/tx/gnd libre 57600 8N1

Mcu serial debug

Connector J8

From left to right (looking at Kello from the front)
tx(rx usb)/rx (tx usb)/ground/empty

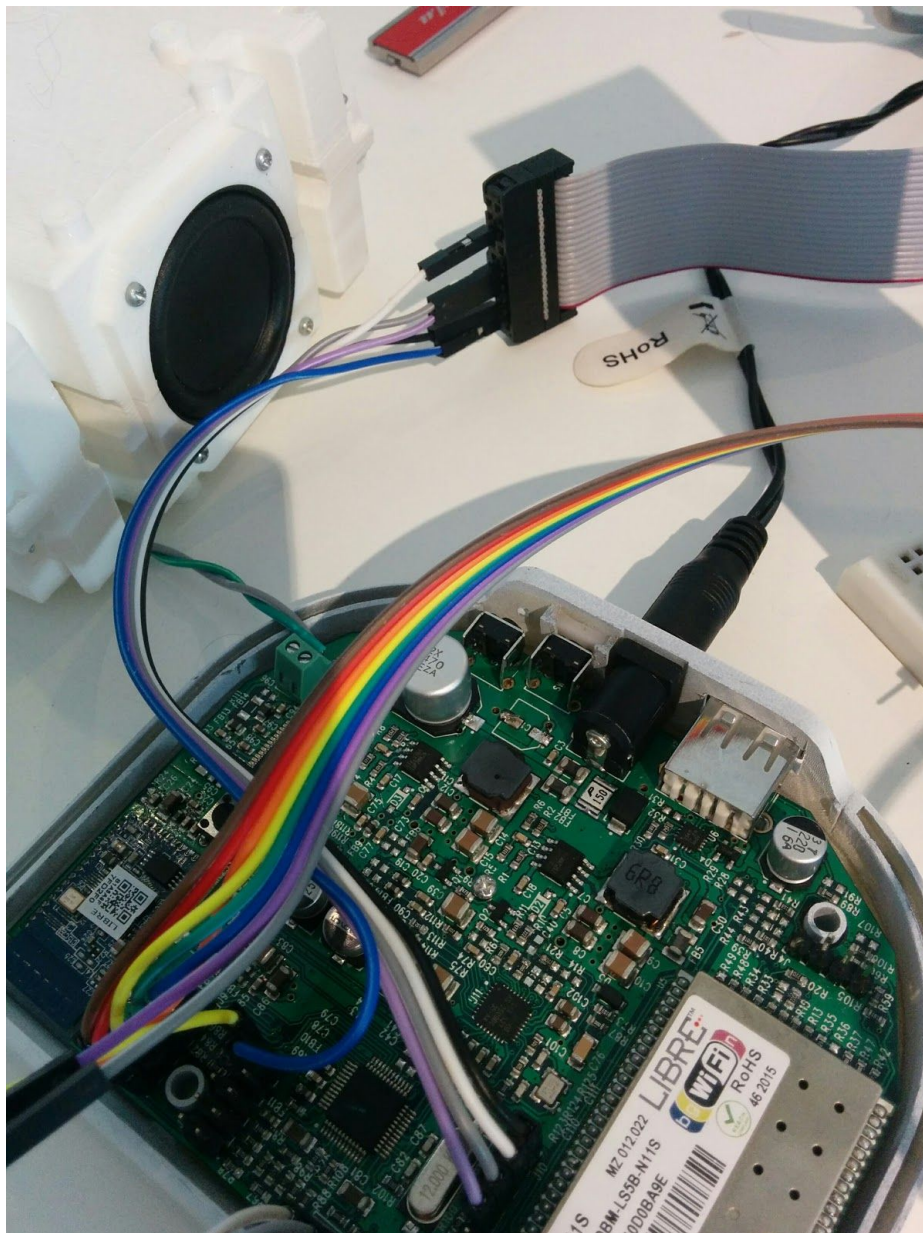
Libre serial debug

Libre connector: J4

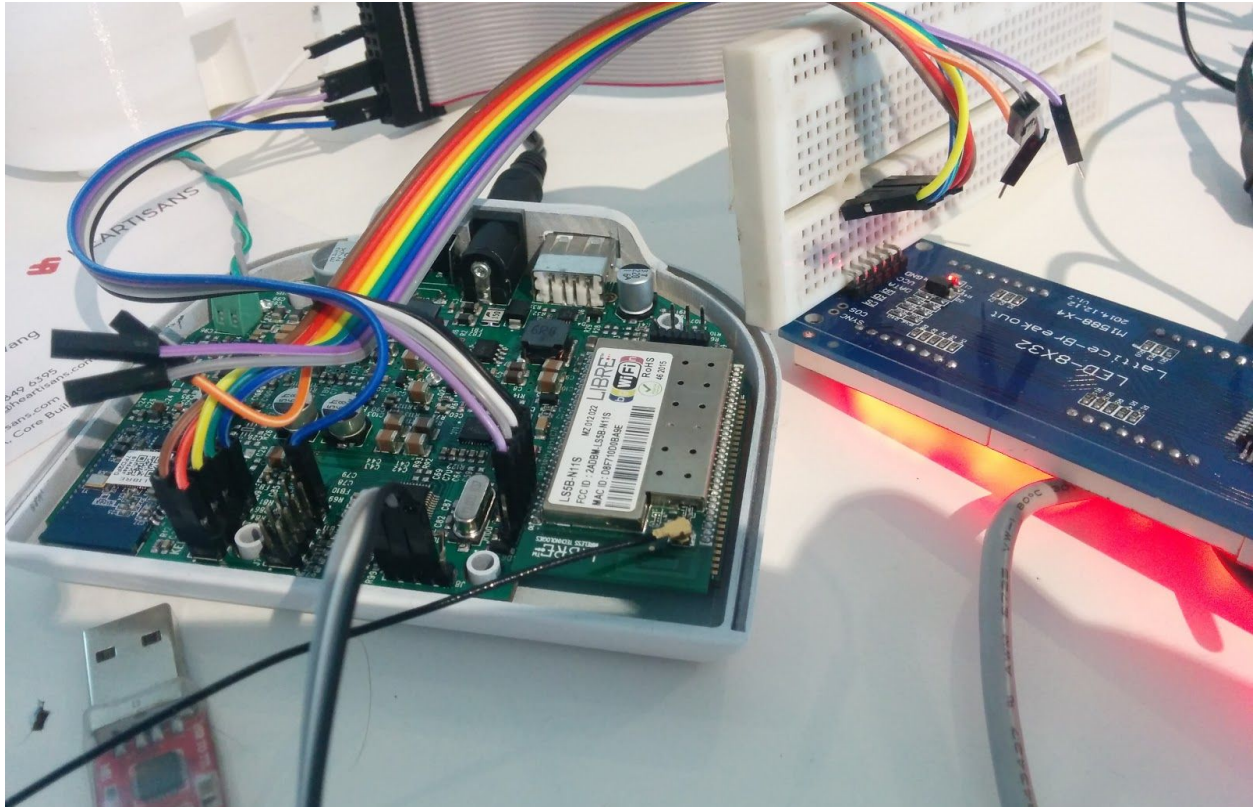
From left to right (looking at Kello from the front):
empty/ground/rx(=tx on usb key)/tx(=rx on usb key)
Baud rate: 57600 8N1

black/white/grey/purple/blue: SWT mcu

red/brown/yellow/orange: VDD/GND/i2c_clk/i2c_data ht16k33 screen



Speaker connector



red/brown/yellow/green/blue: VDD/GND/screen_cs/spi_data/spi_clk ht1632 screen

Configuring network settings for kello

Kello needs to know which Wifi network it has to connect too.

For that, one shall use the demo application “Libre” provided for iOS or Android by LibreWireless.

Tunnel API calls - tunnel_cmd.sh

Upgrade procedure

An file (“Isimage”) will be provided, that contains an version of firmware for kello.

This file shall be copied on a usb key formatted with a FAT partition.

It is highly recommended that the USB key contains no partition table, and only has a raw FAT partition.

Start with emergency procedure

While the USB key is inserted, press the big button at the back and plug the kello on power. The MCU enter emergency upgrade mode, and there is no display. After ~3 mins, kello reboot with the new image.

Start with API call

Once the usb key is ready, plugged into Kello, and Kello has been configured on the network, one can launch the upgrade with the following command :

```
sh ./tunnel_cmd.sh upgrade usb
```

Kello will enter upgrade mode. “U XX” shall be displayed on the screen, where U go from 0 to 100, and then reboot into app mode again

Use a serial port

Use a serial port program, here are a few common :

- minicom on linux
- teraterm on windows
- coolterm on mac

see a few tutorial available on internet

<https://learn.sparkfun.com/tutorials/terminal-basics/coolterm-windows-mac-linux>

You then need to select the (generally) usb-to-serial port you use, and select the speed and serial parameter to get the traces

Driver for adafruit serial-to-usb

<https://www.adafruit.com/products/954>

You might need to install the driver for your platform

- OS X : http://www.prolific.com.tw/US/ShowProduct.aspx?p_id=229&pcid=41

Onboarding on wifi

With app

Launch the app, and hopefully you'll be rightfully guided on how to onboard the device on your wifi

Manually

- * connect wifi AP KelloConf_D0B406
- * connect to "http://192.168.43.1"
- * fill information under "Select Your Network", and click save
- * wait 10s, should display "--:--"

Audio

Electronic notes

Codec NPCA110B routing (cf NPCA110B_DS_1.1.pdf page 3):

input

M1 - libre i2s

M4 - bluetooth analog

AudioEngine Processing (DSP) is used

output

M17 output stereo.

A mixing of L/R track is done after the codec in analog.

It would be better to use the I2S path to do all audio testing.

No easy access was populated unfortunately.

(TBC by EE): maybe we can unpopulate Libre module and use the i2S pins to send audio content to codec.

playing content

through libre

- Use dlina server + controller app, like AllConnect. You might need to upload some mp3 on your mobile phone (android)
- Use spotify connect. Needs internet

through bluetooth

Long tap (2s) on back button to enter pairing mode, pair with your phone, and play from your phone

Interactions

Back button :

- One press: enter WAC (wifi config) mode
- Double tap: check OTA upgrade & upgrade
- Long press: enter BT pairing mode
- Press & plug kello: start upgrade from USB key

Reset button

- Press & plug kello: do factory reset